Shayan Edalatmanesh, Ph.D.

Computational Physical Chemistry | DFT | ML Interatomic Potentials | Scientific Software
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Professional Summary

Computational physicist and physical chemist with expertise in DFT and atomistic simulation of molecular, surface and low-dimensional systems and 20 published articles in Q1. Recent focus: combining agentic AI, machine learned interatomic potentials and DFT, building automated, high-throughput simulation workflows. Author of PPSTM, an open-source STM simulation package in Python, used by research groups internationally.

Education

PhD, Physical Chemistry 2018–2023
Palacký University, Czech Republic & Institute of Physics of the Czech Academy of Sciences, Prague, Czech Republic

Thesis: Theoretical Study of Charge Transport in Nanostructures

MSc, Solid State Physics 2014–2017
Iran University of Science and Technology, Tehran, Iran

Thesis: The Effects of Monovacancies on the Transport Properties of Zigzag Graphene Nanoribbons

BSc, Physics 2008–2013
Iran University of Science and Technology, Tehran, Iran

Experience

Postdoctoral Researcher Jan 2025 – Mar 2026
Spanish National Research Council (CSIC), San Sebastián, Spain

- Built screening pipelines (machine learned interatomic potentials + DFT), cutting calculation cost by ∼70%.
- Designed high-throughput DFT–STM simulation workflows (TMDs), producing curated image datasets for AI model training and benchmarking
- Modelled π–d magnetism in metal–organic nanostructures with DFT and model Hamiltonians (Article)

Postdoctoral Researcher Jun 2023 – Dec 2024
Donostia International Physics Center (DIPC), San Sebastián, Spain

- Studied topological phase transitions in open quantum systems using model Hamiltonians (Article)
- Contributed charge-analysis functionality to *sisl*, an open-source tight-binding/DFT post-processing library

PhD, Theoretical Physical Chemistry Sep 2018 – May 2023
Palacký University & Institute of Physics of the Czech Academy of Sciences, Prague, Czech Republic

- Studied topological phase transitions in 1D polyaromatic hydrocarbons & graphene nanoribbons with DFT
- Developed PPSTM python package for STM simulations of periodic materials
- Extended the **Fireball DFT code** (Fortran) with Zak-phase calculations; wrote Python tools for real-space topological analysis, adopted by collaborators
- Co-supervised 1 MSc and 1 BSc student (resulting papers in *ACS Nano*, *Science Advances*)

Visiting Researcher May 2022 – Jul 2022
Universidad Autónoma de Madrid, Spain (with Dr. Jose Ortega Mateo)

- Built DFT models to study molecular motors on metallic surfaces

Visiting Researcher Nov 2019 – Dec 2019
Aalto University, Finland (with Dr. Adam Foster)

- Developed a DFT-based dipole model to explain organic-inorganic 2D hybrid perovskites

Selected Projects

PPSTM – open-source STM simulation package GitHub · Zenodo DOI

- Extended to periodic systems via Bloch's theorem (crystalline surfaces and slabs, not only isolated molecules)
- Added constant-current scanning (3D grid + isosurface extraction) for experimentally comparable 2D/3D current maps, plus automatic structure overlay and standard file I/O

Agentic AI evaluation for ab initio simulation *Independent project*

- Evaluated an AI agent running VASP DFT calculations autonomously, scored on physical correctness, autonomy, error handling, reasoning and efficiency
- Prototyped automated scoring and a hybrid keyword + embedding retrieval pipeline to ground agent answers in VASP documentation

Publications & Recognition

- **20 peer-reviewed publications** (14 as lead theoretical contributor) in *Nature Nanotechnology*, *Nature Chemistry* (×2), *Nature Communications* (×2), *Science Advances*, *ACS Nano*, *Advanced Materials*, *Angewandte Chemie* Full list available here
- Invited chapter, *Elsevier Encyclopedia of Interfacial Chemistry*: "Topological Quantum States in Carbon Nanostructures" (in preparation, 2026)
- **Fischer Scholarship**, *Outstanding International PhD Student of the Year*, awarded twice (Palacký University)
- Certifications: Introduction to Neural Networks and PyTorch (IBM), AI Agents and Agentic AI with Python & Generative AI (Vanderbilt); Claude Code 101 and Claude Code in Action (Claude Academy)

Technical Skills

Methods: DFT, tight-binding model Hamiltonians, molecular dynamics (i-PI), STM/AFM/KPFM simulation
DFT Codes: VASP, FHI-aims, SIESTA, FIREBALL, Gaussian, ORCA
Technical specialties: Magnetism, spin-orbit coupling, QM/MM, GW, topological band theory
ML & automation: MACE (MLIP), high-throughput screening, simulation dataset generation
Programming: Python (PyTorch, NumPy, SciPy, ASE, Matplotlib, Pandas), Fortran, C++
Tools & HPC: Bash, Git, LaTeX, ssh, SLURM, VESTA, VMD
Databases: Materials Project API, NOMAD

Languages and Interests

Languages: English (fluent), Persian (native), Spanish (B1) and Czech (A2)
Personality traits: Persistent, team player, patient, cheerful, leadership, autonomous, organized
Interests: Football, photography, specialty coffee (former barista champion), cinema, math rock